INCH-POUND
MIL-PRF-32383/7
31 March 2022

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED, SMALL TACTICAL UNIVERSAL BATTERY (STUB), BB-251X/U and BB-252X/U

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-32383.

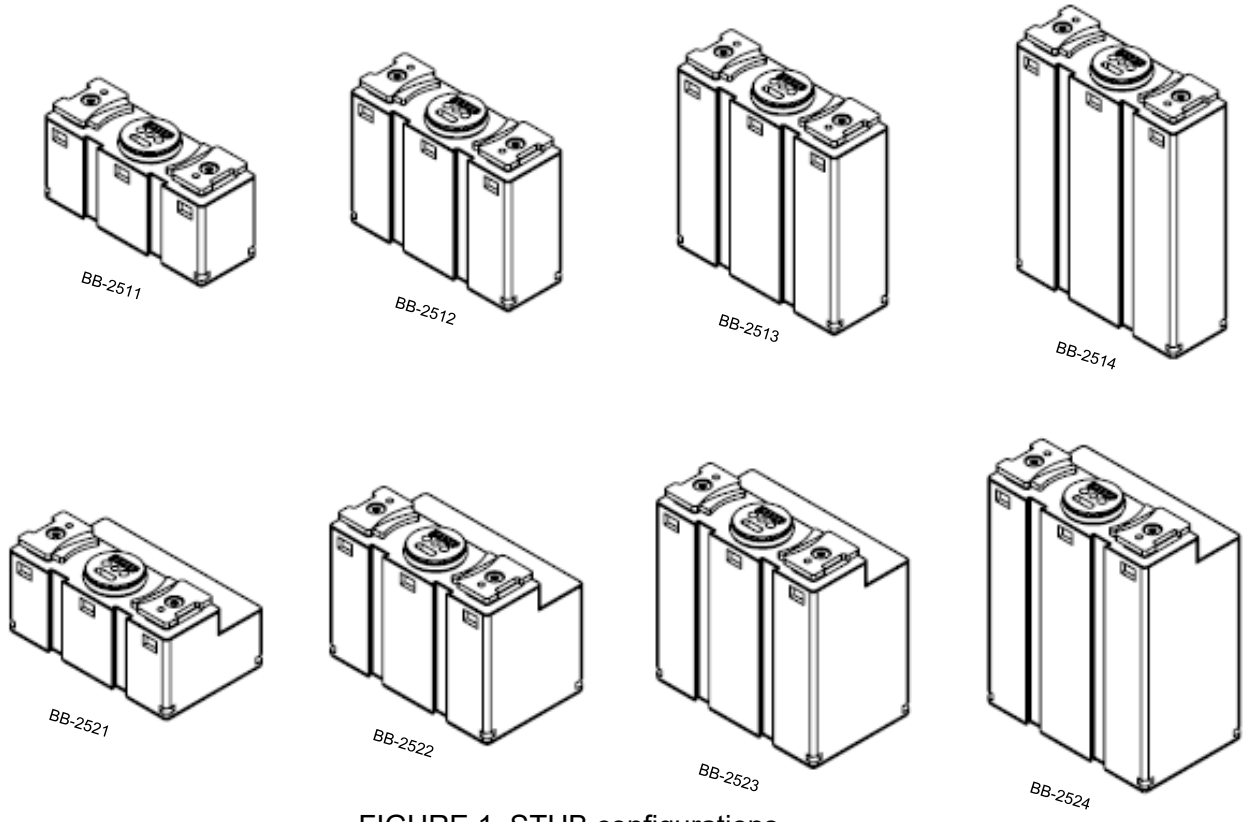


FIGURE 1. STUB configurations

FIGURE 1 NOTE:
Approximate sizes:
BB-2511 2.85 in x 1.00 in x 1.465 in.
BB-2512 2.85 in x 1.00 in x 2.235 in.
BB-2513 2.85 in x 1.00 in x 2.970 in.
BB-2514 2.85 in x 1.00 in x 3.715 in.
BB-2521 2.85 in x 1.75 in x 1.465 in.
BB-2522 2.85 in x 1.75 in x 2.235 in.
BB-2523 2.85 in x 1.75 in x 2.970 in.
BB-2524 2.85 in x 1.75 in x 3.715 in.
See FIGURES 7-14 for exact dimensions.

Battery	PIN
BB-2511	M32383/7-3-11
BB-2512	M32383/7-3-12
BB-2513	M32383/7-3-13
BB-2514	M32383/7-3-14
BB-2521	M32383/7-3-21
BB-2522	M32383/7-3-22
BB-2523	M32383/7-3-23
BB-2524	M32383/7-3-24

REQUIREMENTS
For PIN (see 1.2 for PIN) designation see table to the right

AMSC FSC 6140
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Specification requirements and test methods: The following requirements and test methods of MIL-PRF-32383, identified therein as “when specified”, are applicable as indicated below:

TABLE i. *"When specified" requirements*

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Additional material requirements	*3.3.7*	*4.7.1.9*	*No*
Connector pull force	*3.4.4.1.1.1*	*Certification*	*No*
Connector insertions	*3.4.4.1.1.2*	*4.7.1.3*	*No*
Multilayer finish	3.4.4.1.3	Certification	Yes
Gold plating	*3.4.4.1.4*	*Certification*	*No*
USB power port connector(s)	3.4.4.1.5	4.7.2.17	Yes
USB type A connector(s)	*3.4.4.1.5.1*	*4.7.2.17*	*No*
USB type C connector(s)	3.4.4.1.5.2	4.7.2.17	Yes
Surface friction	3.4.5.6	4.7.1.8	Yes
Item unique identification (IUID)	3.4.6.1.1	4.7.1.1	Yes
Additional physical characteristics requirements	3.4.8	4.7.1.10	Yes
Inrush current	3.5.5	4.7.2.12	Yes
Charge enable	*3.5.6.2*	*4.7.2.6*	*No*
Display, non-smart, always on	*3.5.8.1.1*	*4.7.2.14.1*	*No*
Display, non-smart, user activated	3.5.8.1.2	4.7.2.14.1	Yes
Display, smart	*3.5.8.2*	*4.7.2.14.2*	*No*
SMBus	*3.5.9.1*	*4.7.2.15.1*	*No*
USB power delivery	3.5.9.2	4.7.2.15.2	Yes
Dual-role power device	3.5.9.2.1	4.7.2.14.2	Yes
Additional electrical requirements	3.5.12	4.7.2.18	Yes
Immersion, shallow	3.6.6	4.7.3.8.1	Yes
Immersion, deep	*3.6.6*	*4.7.3.8.2*	*No*
Transit drop, normal	*3.6.7*	*4.7.3.9.1*	*No*
Transit drop, severe	3.6.7	4.7.3.9.2	Yes
Drop while mated	*3.6.7*	*4.7.3.9.2*	*No*
Impact	*3.6.9*	*4.7.3.11*	*No*
Solar radiation	3.6.10	4.7.3.12	Yes
Salt fog	3.6.11	4.7.3.13	Yes
Electrostatic discharge (ESD)	3.6.12	4.7.3.14	Yes
Chemical resistance	3.6.13	4.7.3.15	Yes
Rapid decompression	3.6.14	4.7.3.16	Yes
Explosive atmosphere	3.6.15	4.7.3.17	Yes
Combined temperature, altitude, and humidity	*3.6.16*	*4.7.3.18*	*No*

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TABLE i. *"When specified" requirements* – Continued

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Additional environmental requirements	*3.6.17*	*4.7.3.19*	*No*
Nail penetration, cell	3.7.1.2	4.7.4.4	Yes
Crush, cell	3.7.1.3	4.7.4.5	Yes
Inter-cell protective devices	*3.7.2.6*	*Certification*	*No*
Redundant safety protection	3.7.2.7	Certification	Yes
Projectile	3.7.2.8	4.7.4.11	Yes
Nail penetration, battery	*3.7.2.9*	*4.7.4.12*	*No*
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5	Yes
Weapons characterization	*3.7.2.11*	*4.7.4.13*	*No*
Additional safety requirements	*3.7.3*	*4.7.4.14*	*No*
Electromagnetic interference	3.8	4.7.7	Yes
Additional transportation requirements	*3.9.1*	*4.7.8*	*No*
Reliability	*3.11*	*4.7.2.4.1*	*No*

Details: The following are details specific to M32383/7-3-11, M32383/7-3-12, M32383/7-3-13, M32383/7-3-14, M32383/7-3-23 and M32383/7-3-24 including those referenced in MIL-PRF-32383 as “specified”.

1. Non-Government publications (2.3).
Add:
STUB Software ICD
(Copies of this document may be obtained from https://exocharge.com/products/stub)

2. Certifications (3.2.1).
Add:

USB certification	3.4.6.9.1

3. Dimensions and weight (3.4.3).

TABLE ii. *Dimensions and weight*

		BB-2511	BB-2512	BB-2513	BB-2514	BB-2521	BB-2522	BB-2523	BB-2524
	Max. weight (lb)	0.27	0.39	0.52	0.64	0.39	0.64	0.89	1.20
Dimensions (in.)	Length	2.85	2.85	2.85	2.85	2.85	2.85	2.85	2.85
	Width	1.00	1.00	1.00	1.00	1.75	1.75	1.75	1.75
	Height	1.47	2.24	2.97	3.72	1.47	2.24	2.97	3.72

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4. Connector (3.4.4.1.1)

   Connector: USB Type-C receptacle (see 2.3, The USB Type-C Connectors and Cable Assemblies Compliance Document v 2.1) and the battery interface contacts (See Figure 2) engaged by these methods of attachment: See Figure 15

   Mating connector: N/A

   Multilayer finish: Flat contacts shall be brass with multilayer finish

   Gold plating: N/A

5. USB power port connector(s) (3.4.4.1.5).

   The battery shall contain one USB Type-C connector. The USB connector shall be located as shown in Figure 2-4.

6. USB type C connector(s) (3.4.4.1.5.1).

   a. See Table ii for Discharge rates

   c. The USB connector shall be located as shown in Figure 2-4.

7. Battery case vent (3.4.5.3).

   Pressure release device shall be located on the battery.

8. Surface friction (3.4.5.6).

Batteries shall slide easily in and out of a pouch. When tested as specified in 4.7.1.8 batteries shall have a coefficient of static friction of not greater than 0.40 against a Cordura® fabric containing meta-aramid, nylon and cotton in a blend ratio of 43/30/27 percent by weight.

9. Item unique identification (IUID) (3.4.6.1.1)

   For location, see Figure 3 and 4

Delete:

10. Terminal marking (3.4.6.4).

11. Instructions / notes (3.4.6.5).

    **INSERT preferred charging method, alternate charging method, SOCI matrix, self-discharge rate, maintenance if applicable and other notes if specified. Instructions shall be as brief as clarity allows. (Minimize wording as much as possible.)*

    Preferred charging method: USB PD certified charger or Advanced Battery Charger (ABC) 6130-01-660-3696

    Tap on the side of the battery twice to activate SOCI.

12. Simplified battery label (3.4.6.6).

    The simplified battery label location shall be located on the same side as the charging instructions. See Figure 3 and 4.

13. USB certification marking (3.4.6.9.1)

    Batteries shall be marked with the certified USB marking.

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14. Battery condition for shipping (3.4.7).
Instruction card shall include: safety, storage information, storage temperature, operating temperature, how to activate LEDs, attachment methods, chargers to use, STUB family with energy/dimensions/power delivery, and STUB to STUB charging information.

15. Battery voltage (3.5.2).
Batteries shall operate within the ranges of:
BB-2511, BB-2512, BB-2513, BB-2514, BB-2521, BB-2522, BB-2523, and BB-2524: 3.3-21 V

16. Capacity (3.5.3).
Cut-off voltage: 2.5 V per cell in series
Each battery, discharged at the specified rate to cut-off voltage, shall have the following minimum capacity:

TABLE iii. *Standard/high rate discharge*

Discharge Rate(W)	BB-2511	BB-2512	BB-2513	BB-2514	BB-2521	BB-2522	BB-2523	BB-2524
Standard (W)	15 (5V/3A)	15 (5V/3A)	15 (5V/3A)	15 (5V/3A)	15 (5V/3A)	15 (5V/3A)	15 (5V/3A)	15 (5V/3A)
High Rate (W)	15	24	36	45	24	45	60	80

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TABLE iv. *Capacity table*

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Energy (Wh)					
				BB-2511	BB-2512 & BB-2521	BB-2513	BB-2514 & BB-2422	BB-2523	BB-2524
Capacity Discharge	3.5.3	4.7.2.3	Table iii	10	21.25	32.5	42	63	84
Cycle life (224th cycle)	3.5.4	4.7.2.4	Table iii	8.8	17.6	19.2	25.6	38.4	68.8
High rate discharge	3.5.3	4.7.2.5	Table iii	8.75	13.2	31	22.5	45	40
Low temperature discharge	3.5.3	4.7.2.7	Table iii	8	16.8	24.8	33.6	50.4	67.2
High temperature discharge	3.5.3	4.7.2.8	Table iii	9.5	20	29.5	39.9	59.9	79.8
Inrush current	3.5.5	4.7.2.12	See 3.5.5	8	16.8	24.8	33.6	50.4	67.2
Charge acceptance	3.5.10.1	4.7.2.9	Table iii	10	21	31	42	63	84
High temperature charge acceptance	3.5.10.1	4.7.2.9	Table iii	8.8	18.5	27.3	37	55.4	73.9
Low temperature charge acceptance	3.5.10.1	4.7.2.9	Table iii	8.8	18.5	27.3	37	55.4	73.9
Retention of charge	3.5.10.2	4.7.2.10	Table iii	9.9	20.8	30.7	41.6	62.4	83.2
Pulse discharge	3.5.3	4.7.2.11	See 4.7.2.11	8.8	18.5	27.3	37	55.4	73.9
Extreme low temperature discharge	3.6.1	4.7.3.2	Table iii	7	14.7	21.7	29.4	44.1	58.8
Extreme high temperature discharge	3.6.1	4.7.3.3	Table iii	9.5	20	29.5	39.9	59.9	79.8
Temperature shock (post-test)	3.6.3	4.7.3.5	Table iii	8	16.8	24.8	33.6	50.4	67.2
Mechanical shock (post-test)	3.6.3	4.7.3.6	Table iii	10	21	31	42	63	84
Vibration (during test)	3.6.5	4.7.3.7	Table iii	10	21	31	42	63	84
Battery storage life	3.6.8	4.7.3.10	Table iii	9	18.9	27.9	37.8	56.7	75.6
Overcharge/electric leakage (post-test)	3.7.2.2	4.7.4.7	Table iii	10	21	31	42	63	84
Short circuit protection (post-test)	3.7.2.3	4.7.4.8	Table iii	10	21	31	42	63	84
High temperature temporary cut off	3.7.2.4	4.7.4.9	Table iii	9.5	20	29.5	39.9	59.9	79.8
Solar Radiation	3.6.10	4.7.3.12	Table iii	10	21	31	42	63	84
Salt Fog	3.6.11	4.7.3.13	Table iii	10	21	31	42	63	84

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TABLE iv. *Capacity table* - Continued

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Energy (Wh)					
				BB-2511	BB-2512 & BB-2521	BB-2513	BB-2514 & BB-2422	BB-2523	BB-2524
ESD	3.6.12	4.7.3.14	Table iii	10	21	31	42	63	84
Rapid decompression	3.6.14	4.7.3.16	Table iii	10	21	31	42	63	84
Explosive Atmosphere	3.6.15	4.7.3.17	Table iii	10	21	31	42	63	84

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17. Cycle Life (3.5.4)
Battery shall be capable of performing 224 cycles

18. Inrush current (3.5.5)
The discharge profile shall be no less than as follows.

Current (A)	Over Current Protection (OCP) trip peak current (A)	OCP trip time (milliseconds)
5.00	6.00	10
4.00	4.80	10
3.00	3.60	10
2.00	2.40	10
1.67	2.00	10
1.50	1.80	10
1.25	1.50	10
1.13	1.48	10
1.00	1.20	10
0.75	0.90	10

19. Charging interface (3.5.6.1).
Battery shall have two charging interfaces; (1) USB Type C receptacle, and (2) interface described in Figure 2.

20. Additional electrical requirements (3.5.12).
Batteries shall comply with unstructured vendor defined message (UVDM) requirements as specified in the STUB Software ICD at exocharge.com/products/stub

21. High temperature temporary cut off devices (3.7.2.4).
Not less than two devices

22. High temperature permanent cut off devices (3.7.2.5).
Not less than two devices

23. Redundant safety protection (3.7.2.7).
Batteries shall have a battery management system, no less than two FETS, a resettable and permanent fuse protecting all of the terminals and external connections. All three tiers of protection shall have had to fail prior to allowing any potential uncontrolled power flow to the battery cells that would cause an unsafe condition from any signal or power pins on any connector ports.

VERIFICATION

24. First article inspection (4.4).
e. Six samples

25. **Replace:**
Table X with the following:

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TABLE v. *First articles tests, complete batteries*

			Samples or Groups (GR)															
			GR 1										GR 2	GR 3	GR 4	GR 5	GR 6	GR 7
Inspection/Test	**Req't Paragraph**	**Test Method Paragraph**	**1**	**2**	**3**	**4**	**5, 6**	**7, 8**	**9, 10**	**11, 12**	**13, 14**	**15, 16**	**(5 ea)**	**(8 ea)**	**(12 ea)**	**1-6**	**7-18**	**(10 ea)**
Visual and mechanical	3.4	4.7.1.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	☑	✓	☑	☑	☑	☑
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓	✓	✓	□	□	□	□	□	□	□	☑	☑	☑
End item interoperability	3.10.2	4.7.6.2	☑	☑	□	□	□	□	□	□	□	□	□	□	□	□	□	☑
Battery open circuit voltage	3.5.2	4.7.2.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	☑	✓	✓	☑	☑	☑
Insulation resistance	3.7.2.1	4.7.4.6	✓	✓	✓	✓	✓	✓	✓	✓	□	□	□	□	□	□	□	□
Altitude	3.6.2	4.7.3.4	✓	✓	✓	✓	✓	✓	✓	✓	□	□	□	□	□	□	□	□
Charge acceptance	3.5.10.1	4.7.2.9	□	□	□	□	□	□	□	✓	□	□	□	□	□	□	□	□
Concurrent charge/discharge	3.5.11	4.7.2.16	□	□	□	□	□	□	☑	□	□	□	□	□	□	□	□	□
Capacity discharge (initial)	3.5.3	4.7.2.3.1	✓	✓	✓	✓	✓	✓	✓	✓	□	□	□	□	□	☑	☑	☑
Cycle life	3.5.4	4.7.2.4	□	□	□	□	✓	□	□	□	□	□	□	□	□	□	□	□
Battery storage life	3.6.8	4.7.3.10	□	□	□	□	□	✓	□	□	□	□	□	□	□	□	□	□
Overcharge / electrical leakage	3.7.2.2	4.7.4.7	□	□	□	□	□	□	✓	□	□	□	□	□	□	□	□	□
Low temperature discharge	3.6.1	4.7.2.7	□	□	□	□	□	□	✓	□	□	□	□	□	□	□	□	□
High temperature discharge	3.6.1	4.7.2.8	□	□	□	□	□	□	✓	□	□	□	□	□	□	□	□	□
Projectile	3.7.2.8	4.7.4.11	□	□	□	□	□	□	☑	□	□	□	□	□	□	□	□	□
Nail Penetration, battery	3.7.2.9	4.7.4.12	□	□	□	□	□	□	□	□	□	□	□	□	□	☑	□	□
High rate discharge	3.5.3	4.7.2.5	✓	□	✓	□	□	□	□	□	□	□	□	□	□	□	□	□
Retention of charge	3.5.10.2	4.7.2.10	✓	✓	✓	✓	□	□	□	□	□	□	□	□	□	□	□	□
Pulse discharge	3.5.3	4.7.2.11	✓	□	✓	□	□	□	□	□	□	□	□	□	□	□	□	□
Inrush current	3.5.5	4.7.2.12	□	□	□	□	□	□	□	☑	□	□	□	□	□	□	□	□
Temperature shock	3.6.3	4.7.3.5	✓	✓	✓	✓	□	□	□	□	□	□	□	□	□	□	□	□
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓	□	□	□	□	□	□	□	□	□	□	□	□

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TABLE v. *First articles tests, complete batteries* – Continued

Vibration (during test)	3.6.5	4.7.3.7	✓	✓	✓	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Vibration (loose cargo)	3.6.5	4.7.3.7.1	☑	☑	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Chemical resistance	3.6.13	4.7.3.15	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Connector insertion	3.4.4.1.1.2	4.7.1.3	✓	☐	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Flat terminal strength	3.4.4.1.2	4.7.1.4	✓	☐	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
USB power port connector(s)	3.4.4.1.5	4.7.2.17	☑	☑	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Immersion, shallow	3.6.6	4.7.3.8.1	☑	☑	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Immersion, deep	3.6.6	4.7.3.8.2	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐
Transit drop, normal	3.6.7	4.7.3.9.1	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐	☐
Transit drop, severe	3.6.7	4.7.3.9.2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Surface friction	3.4.5.6	4.7.1.8	☐	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Reliability	3.11	4.7.2.4.1	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐	☑
Impact	3.6.9	4.7.3.11	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐
Salt fog	3.6.11	4.7.3.13	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐	☐
Electrostatic discharge (ESD)	3.6.12	4.7.3.14	☑	☑	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Solar radiation	3.6.10	4.7.3.12	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Rapid decompression	3.6.14	4.7.3.16	☑	☑	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Explosive atmosphere	3.6.15	4.7.3.17	☐	☑	☑	☐	☐	☐	☐	☑	☑	☑	☐	☑	☐	☐	☐	☐
Combined temperature, humidity, and altitude	3.6.16	4.7.3.18	☐	☐	☐	☐	☐	☐	☐	✓	☐	☐	☐	☐	☐	☐	☐	☐
Immersion, shallow (post drop)	3.6.6	4.7.3.8.1	☑	☑	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Immersion, deep (post drop)	3.6.6	4.7.3.8.2	✓	✓	✓	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
State of charge indicator	3.5.8	4.7.2.14	☐	☐	☐	☐	☑	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐
SMBus	3.5.9.1	4.7.2.15.1	☐	☑	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Full capacity discharge	3.5.3	4.7.2.3	☑	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Concurrent charge / discharge	3.5.11	4.7.2.16	✓	☐	☐	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

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TABLE v. *First articles tests, complete batteries* – Continued

Extreme low temp. discharge	3.6.1	4.7.3.2	☐	✓	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Extreme high temp. discharge	3.6.1	4.7.3.3	☐	✓	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Battery case vent	3.4.5.3	4.7.1.6	☐	✓	✓	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Short circuit protection	3.7.2.3	4.7.4.8	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐
High temp. temporary cut-off	3.7.2.4	4.7.4.9	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	✓	☐	☐	☐	☐
High temp. permanent cut-off	3.7.2.5	4.7.4.10	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐
Electromagnetic interference	3.8	4.7.7	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐
Interoperability	3.1	4.7.6	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐
Weapons characterization	3.7.2.11	4.7.4.13	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

26. **Delete:**
    4.6.1 Constant potential, current limited.
    Use alternate charging (4.6.3) when Constant potential, current limited is required

27. Alternate charging (4.6.3).
    1. *USB certified PD Charger*
    2. *USB certified PPS Charger*

28. Cycle life test (4.7.2.4).

a. and b.

BB-2511	15 W
BB-2512	15 W
BB-2513	15 W
BB-2514	30 W
BB-2521	30 W
BB-2522	30 W
BB-2523	30 W
BB-2524	30 W

c. 224 Cycles

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29. Pulse discharge (4.7.2.11).
    d. Discharge with the battery at values below for 5 seconds, rest for 25 seconds.

BB-2511	15 W
BB-2512	30 W
BB-2513	30 W
BB-2514	30 W
BB-2521	60 W
BB-2522	60 W
BB-2523	60 W
BB-2524	60 W

30. Immersion, shallow (4.7.3.8.1).
    Configuration 1: Stand alone, not connected to any equipment
    Configuration 2: Connected to specified equipment
    After all batteries have been subjected to configuration 1 testing, 50% of the batteries shall be subjected to configuration 2 with Twist method of attachment and 50% of the batteries shall be subjected to configuration 2 with Direct Insert or Hook & Snap method of attachment.
    a. For configuration 1: 6 feet for 30 minutes
       For configuration 2: 9 feet for 15 minutes

Note: Configuration 2 is only applicable when o-ring is engaged (Twist, Direct Insert and Hook & Snap) See Figure 15.

31. Nail penetration, cell (4.7.4.4).
    SOC not less than 50%

32. Projectile (4.7.4.11).
    SOC not less than 50%

33. Nail penetration, battery (4.7.4.12).
    SOC not less than 50%

34. Electromagnetic interference (4.7.7).

The battery EMI shall be characterized with an approved U.S. Army system (radio or similar) defined and provided at contract award as a load for the battery. Testing shall be conducted in accordance with associated test procedures of MIL-STD-461. Conducted tests shall reflect battery contributions to MIL-STD-461 requirements and shall be recorded and verified to not exceed that of MIL-STD-461 limits.
Note, the Government may require personnel with security clearance or may conduct independent testing and adjudication of independent test results.

There are two possible configurations for Electromagnetic Interference (EMI) Testing:

b. Configuration 1 (ref: MIL-STD-461, FIGURE 2) has the battery under test attached to a USB PD electrical compliance test unit with source & sink capabilities for various PDOs, as well as PPS capability. In this configuration, the USB PD electrical compliance test unit represents the power source.

c. Configuration 2 (ref: MIL-STD-461, FIGURE 2) has the battery under test attached to and powering a specified Army equipment.

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CE106, conducted emissions, antenna port: N/A

CS103, conducted susceptibility, antenna port, intermodulation: N/A

CS104, conducted susceptibility, antenna port, rejection of undesired signals: N/A

CS105, conducted susceptibility, antenna port, cross modulation: N/A

CS109, conducted susceptibility, structure current: N/A

CS114, conducted susceptibility, bulk cable injection: Testing shall utilize AIRCRAFT (EXTERNAL OR SAFETY CRITICAL) level limits of TABLE VI, CS114 limit curves for Army platforms. The injection and monitor probes shall be places on the type C cable bundle including signal and ground, 5 cm from the EUT. Verify by test using configuration #1 that the battery complies CS114-1 figure limits for the highest source and sink PDO. For batteries with PPS sink support, verify by test using configuration #1 that the battery complies with CS114-1 figure limits for the battery's requested charging voltage. For batteries with PPS source support, verify by test using configuration #1 that the battery complies with CS114-1 figure limits for the radio's nominal input voltage.

CS115, conducted susceptibility, bulk cable injection, impulse excitation: Verify by test using configuration #1 that the battery exhibits no performance degradation for the highest source and sink PDO. For batteries with PPS sink support, verify by test using configuration #1 that the battery exhibits no performance degradation for the battery's requested charging voltage. For batteries with PPS source support, verify by test using configuration #2 that the battery exhibits no performance degradation when powering the radio at the radio's nominal input voltage.

CS116, conducted susceptibility, damped sinusoidal transients, cables and power leads: Use limits of FIGURE CS116-2. Verify by test using configuration #1 that the battery complies CS116-2 figure limits for the highest source and sink PDO. For batteries with PPS sink support, verify by test using configuration #1 that the battery complies with CS116-2 figure limits for the battery's requested charging voltage. For batteries with PPS source support, verify by test using configuration #2 that the battery complies with CS116-2 figure limits for the radio's nominal input voltage.

CS117, conducted susceptibility, lightning induced transients, cables and power leads: N/A

CS118, personnel borne electrostatic discharge: Test points shall include, at a minimum, the 1) battery housing 2) any Vbatt, GND, and SMBus clk/data pads 3) Vbus and GND for the type-C connector. Verify by test using configuration #1 that the battery exhibits no performance degradation for the highest source and sink PDO. For batteries with PPS sink support, verify by test using configuration #1 that the battery exhibits no performance degradation for the battery's requested charging voltage. For batteries with PPS source support, verify by test using configuration #2 that the battery exhibits no performance degradation when powering the radio at the radio's nominal input voltage.

RE101, radiated emissions, magnetic field: Use the limits of FIGURE RE 101-1 ("RE101-1. RE101 limit for all Army applications”). Verify by test using configuration #1 that the battery complies RE101-1 figure limits for the highest source and sink PDO. For batteries with PPS sink support, verify by test using configuration #1 that the battery complies with RE101-1 figure limits for the battery's requested charging voltage. For batteries with PPS source support, verify by test using configuration #2 that the battery complies with RE101-1 figure limits for the radio's nominal input voltage.

RE102, radiated emissions, electric field: Use the limits of FIGURE RE 102-4 (“Navy Mobile & Army” limit in Figure "RE102-4. RE102 limit for ground applications”). Verify by test using configuration #1 that the battery complies RE102-4 figure limits for the highest source and sink PDO. For batteries with PPS sink support, verify by test using configuration #1 that the battery complies with RE102-4 figure limits for the battery's requested

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charging voltage. For batteries with PPS source support, verify by test using configuration #2 that the battery complies with RE102-4 figure limits for the radio's nominal input voltage.

RE103, radiated emissions, antenna spurious and harmonic outputs: N/A

RS101, radiated susceptibility, magnetic field: N/A

RS103, radiated susceptibility, electric field: Testing shall utilize “AIRCRAFT (EXTERNAL OR SAFETY CRITICAL)” level limits for 2 MHz to 18 GHz for Army platforms. Test to be conducted in configuration #2. Record thresholds of susceptibility in accordance with MIL-STD-461 paragraph 4.3.10.4.3 and data presentation requirements of 5.21.4.5 and mark all data in accordance with PM provided security guidelines.

RS105: N/A

35. Approved chargers (6.5).

Remove:

TYPE DESIGNATION	COMMON NAME	NIIN	PART NUMBER	CAGEC
PP-8498/U	Soldier Portable Charger (SPC)	01-495-2839	BTC-70801	51828
PP-8481B/U	Vehicle Mounted Charger (VMC)	01-527-2726	BTC-70836	51828

36. STUB nomenclature (6.18.1).

In determining the nomenclatures for these batteries, the government used the following structure:

BB-25XY

BB- Rechargeable
25- Li-Ion Chemistry
X- Battery width reference number
Y- Battery Height reference number

ADDITIONAL FIGURES

(On the following pages)

Note: Dimensions are in inches. Tolerances; if two decimals are present, ± 0.01 inches and if three decimals are present, ± 0.005 inches, unless stated otherwise

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REFERENCE NUMBER	OUTPUT PDOs & APDO
BB-2511	5V/3A
	9V/1.67A
	12V/1.25A
	15V/1A
	20V/0.75A
	3.3V - 5.9V/3A
	3.3V - 11V/1.5A
BB-2512 BB-2521	5V/3A
	9V/2.75A
	12V/2A
	15V/1.5A
	20V/1.125A
	3.3V - 11V/3A
	3.3V - 21V/1.5A
BB-2513	5V/3A
	9V/3A
	12V/3A
	15V/2A
	20V/1.5A
	3.3V - 11V/3A
	3.3V - 21V/2A
BB-2514 BB-2522	5V/3A
	9V/3A
	12V/3A
	15V/3A
	20V/2A
	3.3V - 16V/3A
	3.3V - 21V/2A
BB-2523	5V/5A
	9V/5A
	12V/5A
	15V/4A
	20V/3A
	3.3V - 16V/4A
	3.3V - 21V/3A
BB-2524	5V/5A
	9V/5A
	12V/5A
	15V/5A
	20V/4A
	3.3V - 16V/5A
	3.3V - 21V/4A

TABLE 2

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ELECTRICAL CONNECTIONS

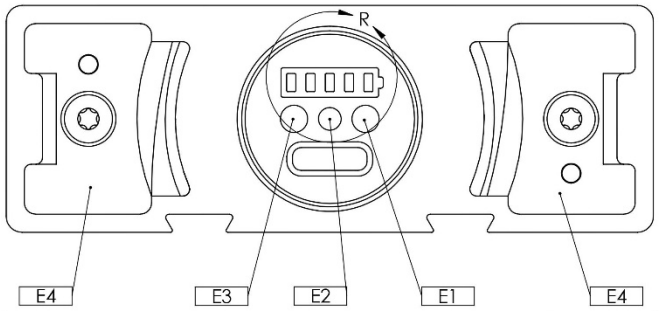
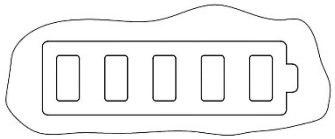


FIGURE A



DETAIL R
SCALE 5 : 1

CONTACT REF DES	CONNECTION	MATERIAL	FINISH
E1	VBUS	304/316 SS	PASSIVATE
E2	CC		
E3	VBUS		
E4	GND	316 SS	PASSIVATE

TABLE 3

NOTES: (UNLESS OTHERWISE SPECIFIED)

1. VBUS HAS NO OUTPUT UNTIL CC1 CONNECTION IS PRESENT PER USB POWER DELIVERY SPECIFICATION REV 3.0 VER 2.0.

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FIGURE 2. BB-25XX()/U electrical connections

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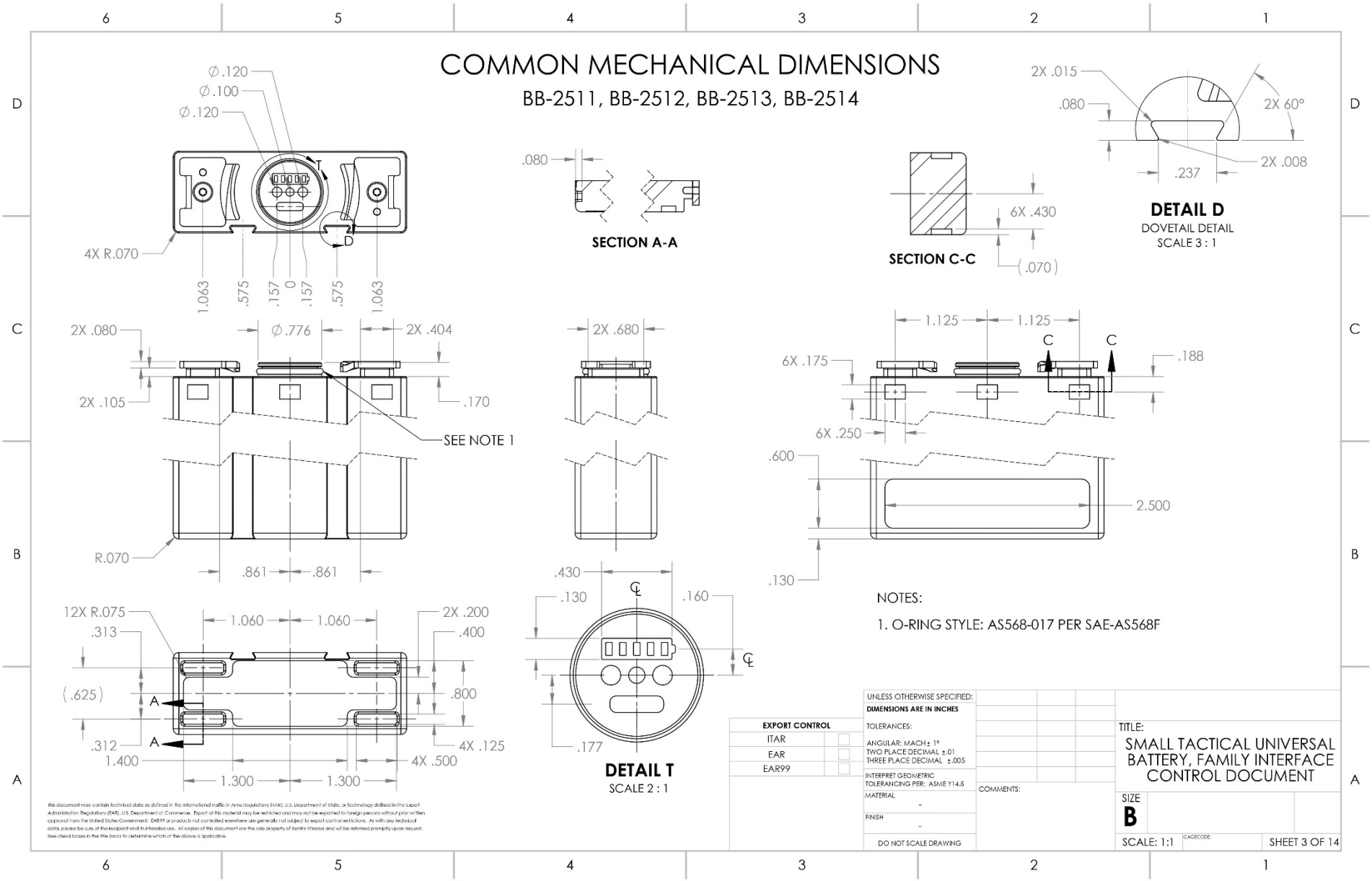


FIGURE 3. BB-251X()/U common mechanical dimensions

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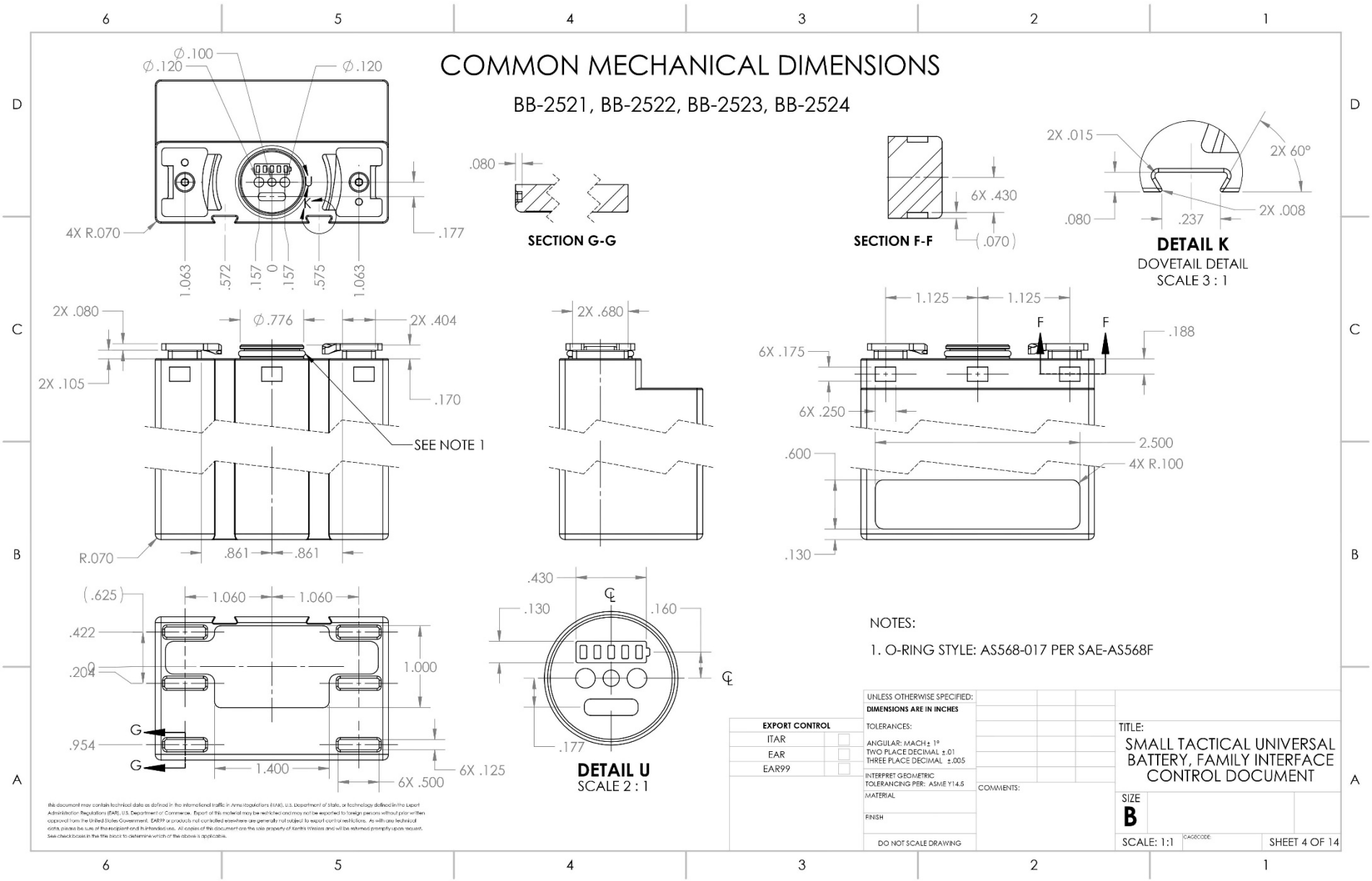


FIGURE 4. BB-252X()/U common mechanical dimensions

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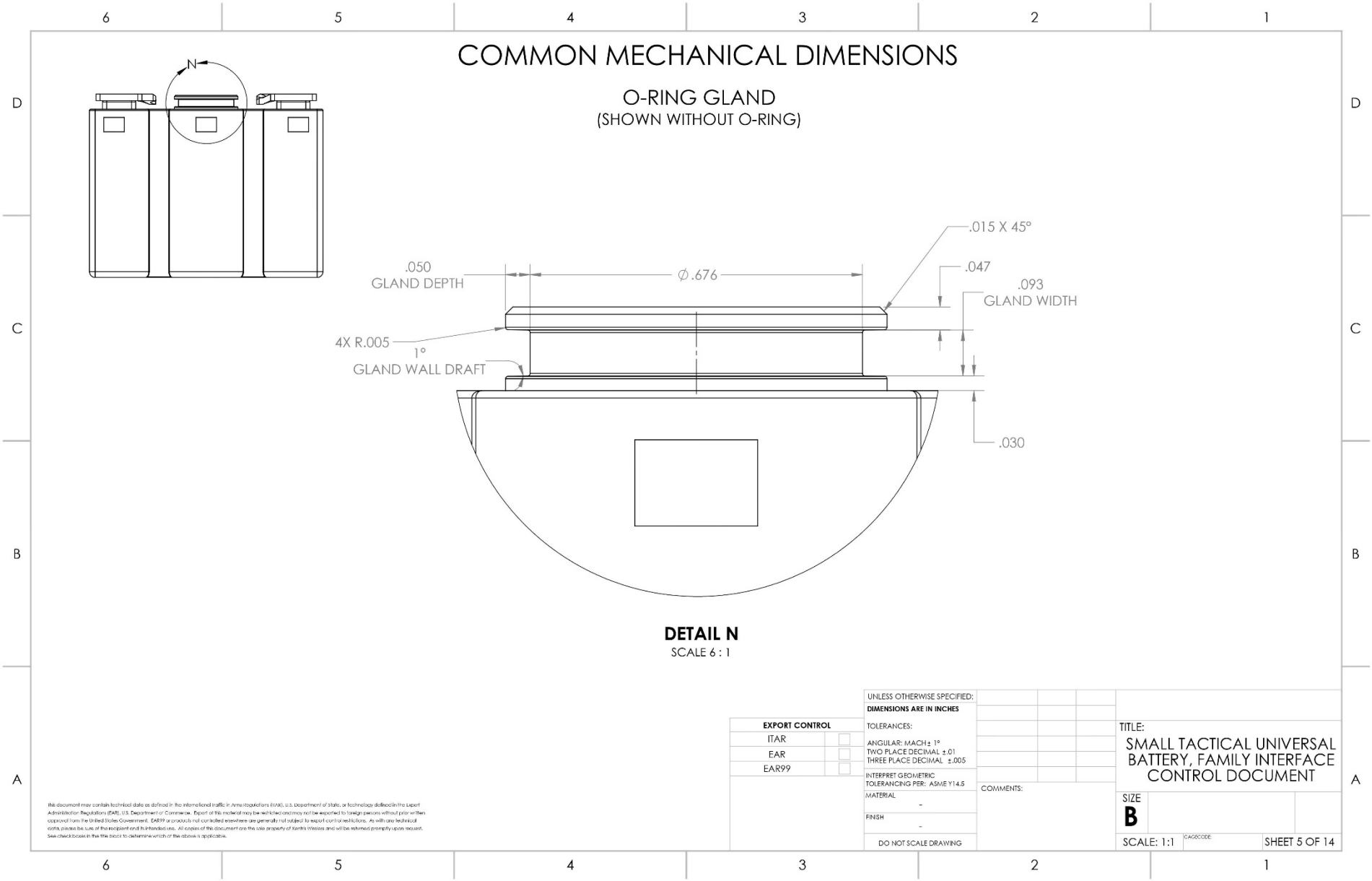


FIGURE 5. BB-25XX()/U O-ring gland

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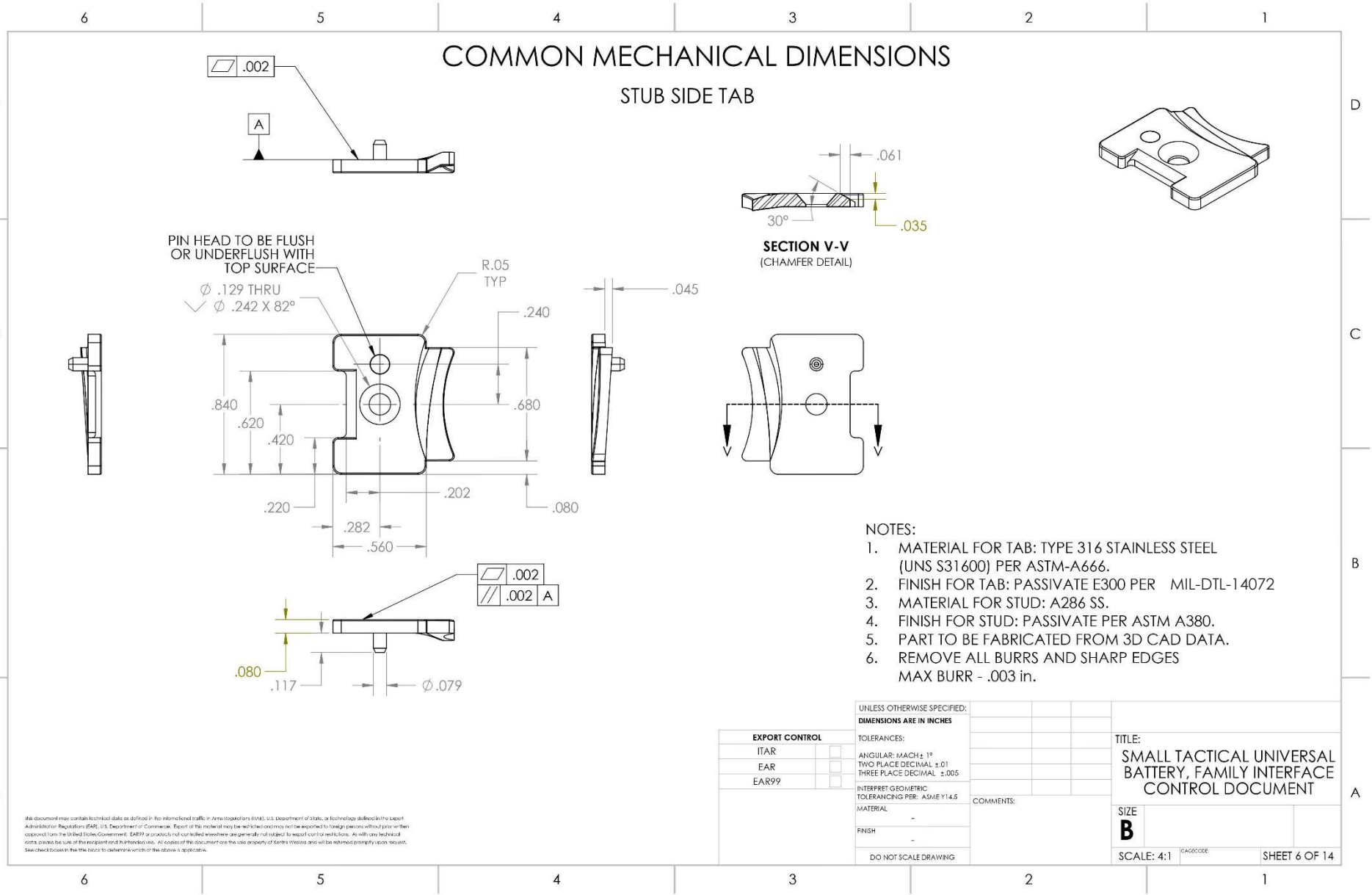


FIGURE 6. BB-25XX()/U side tab

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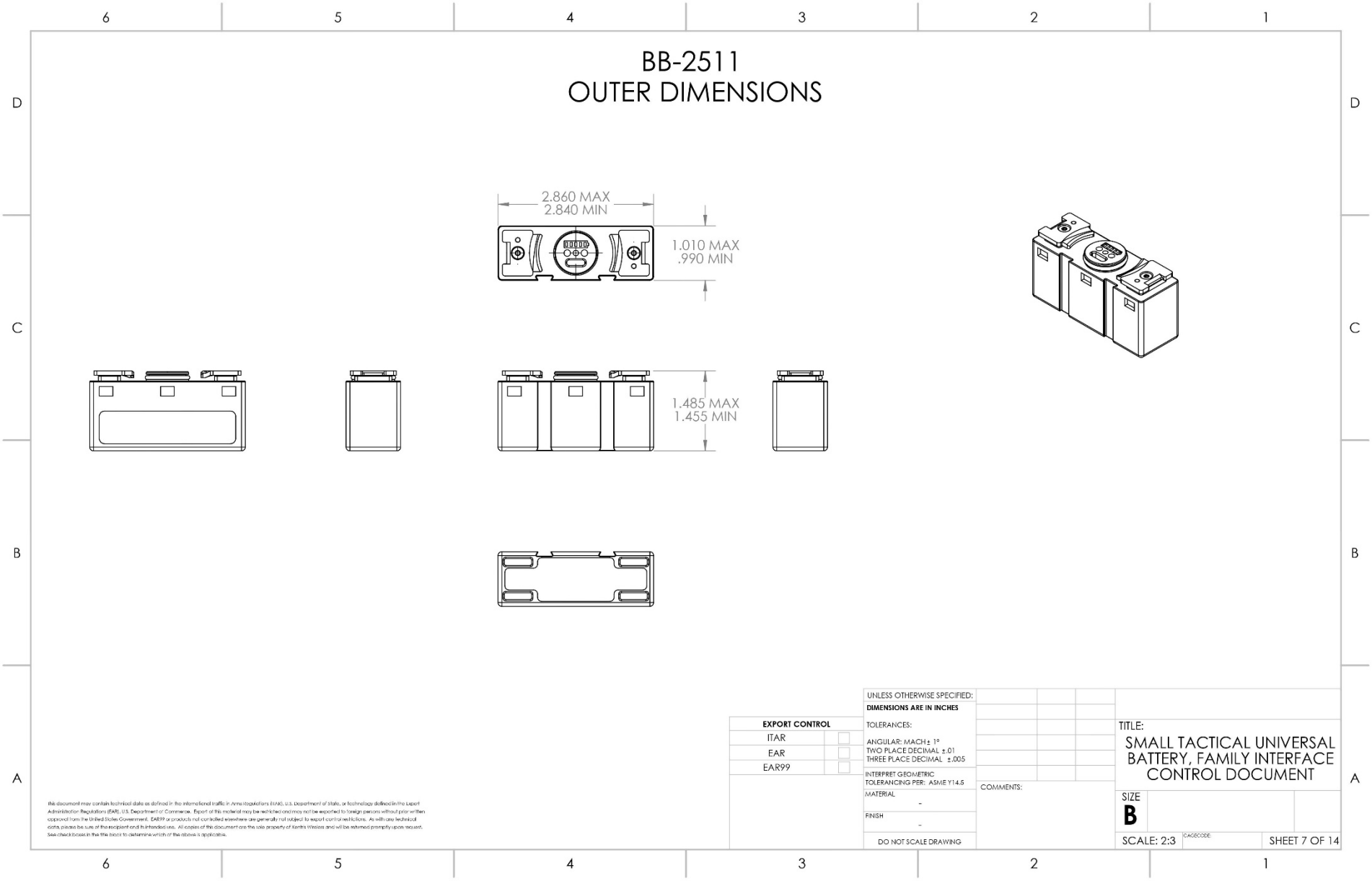
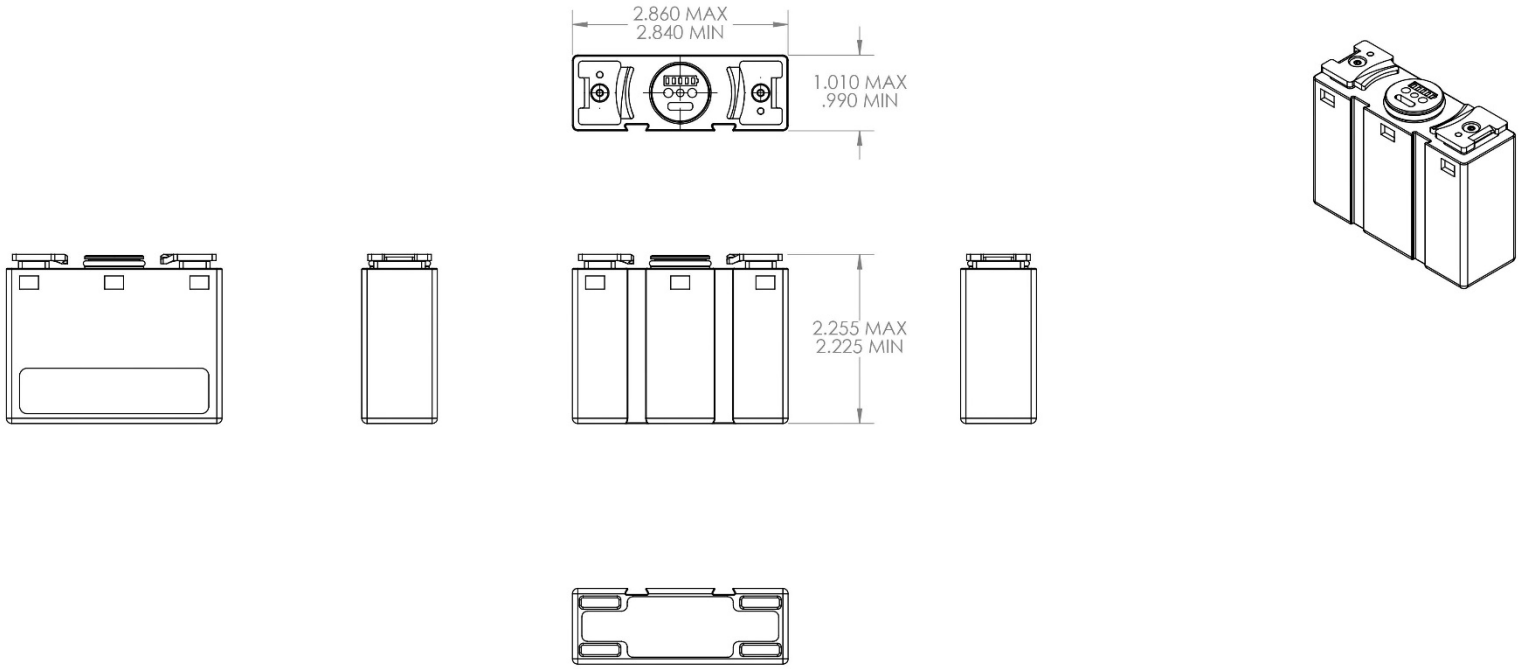


FIGURE 7. BB-2511/U

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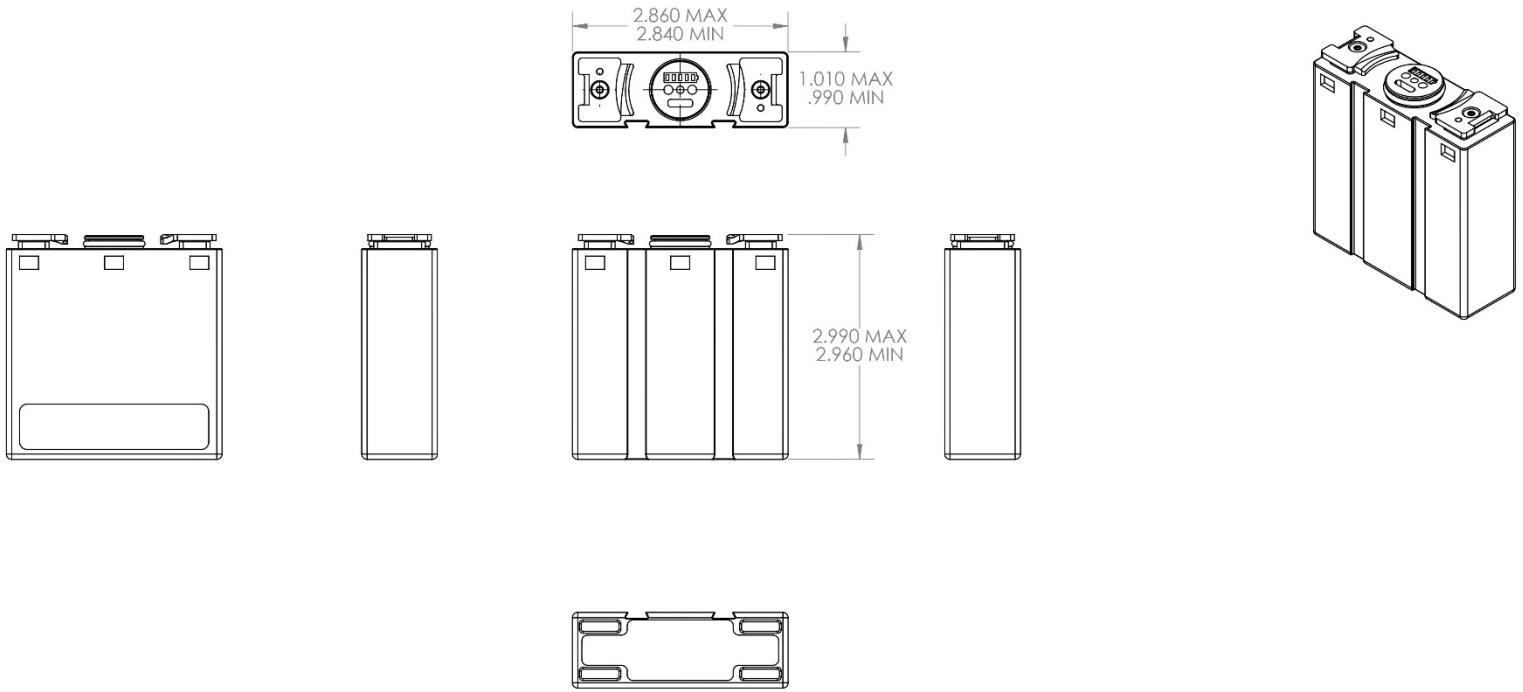
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FIGURE 8. BB-2512/U

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THREE PLACE DECIMAL ±.005
INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5
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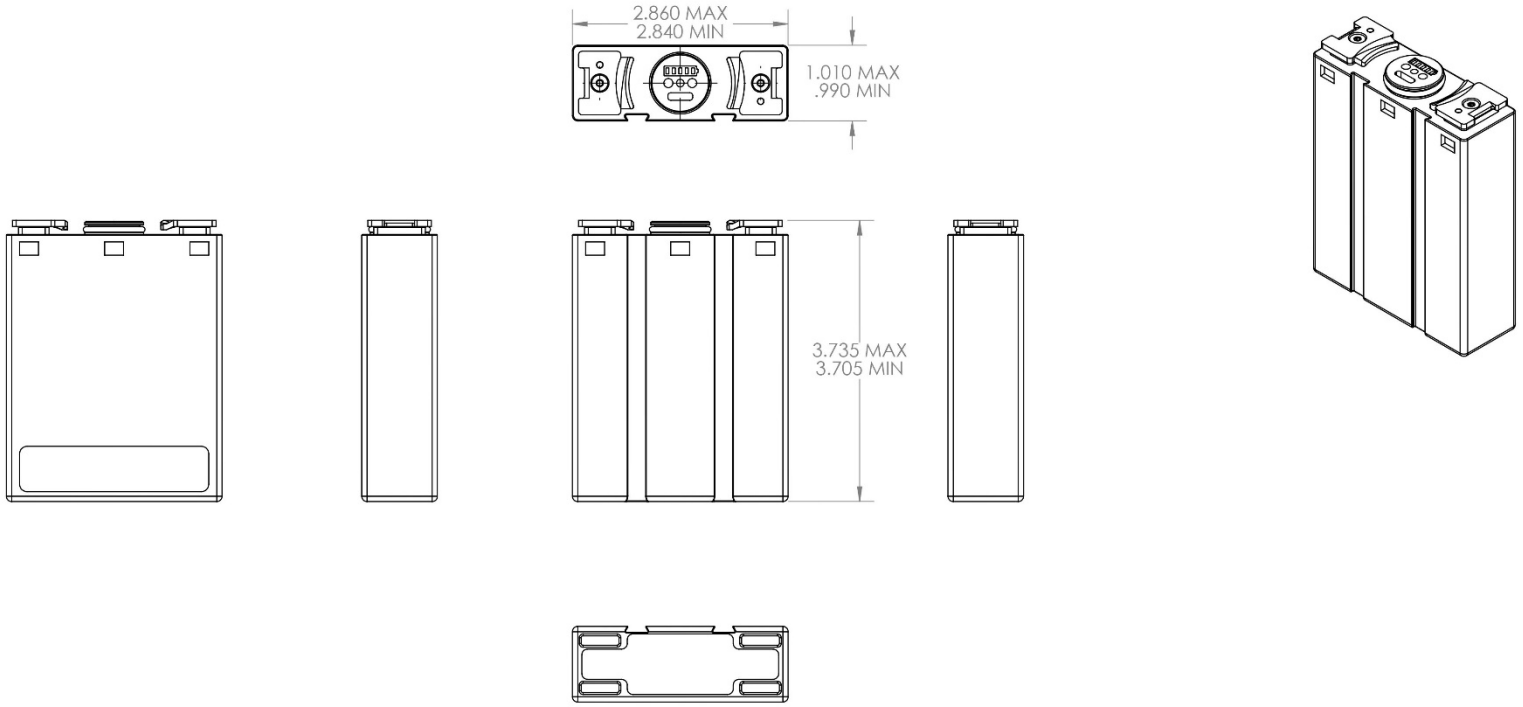
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FIGURE 9. BB-2513/U

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EAR	☐
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THREE PLACE DECIMAL ±.005
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MATERIAL -
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FIGURE 10. BB-2514/U

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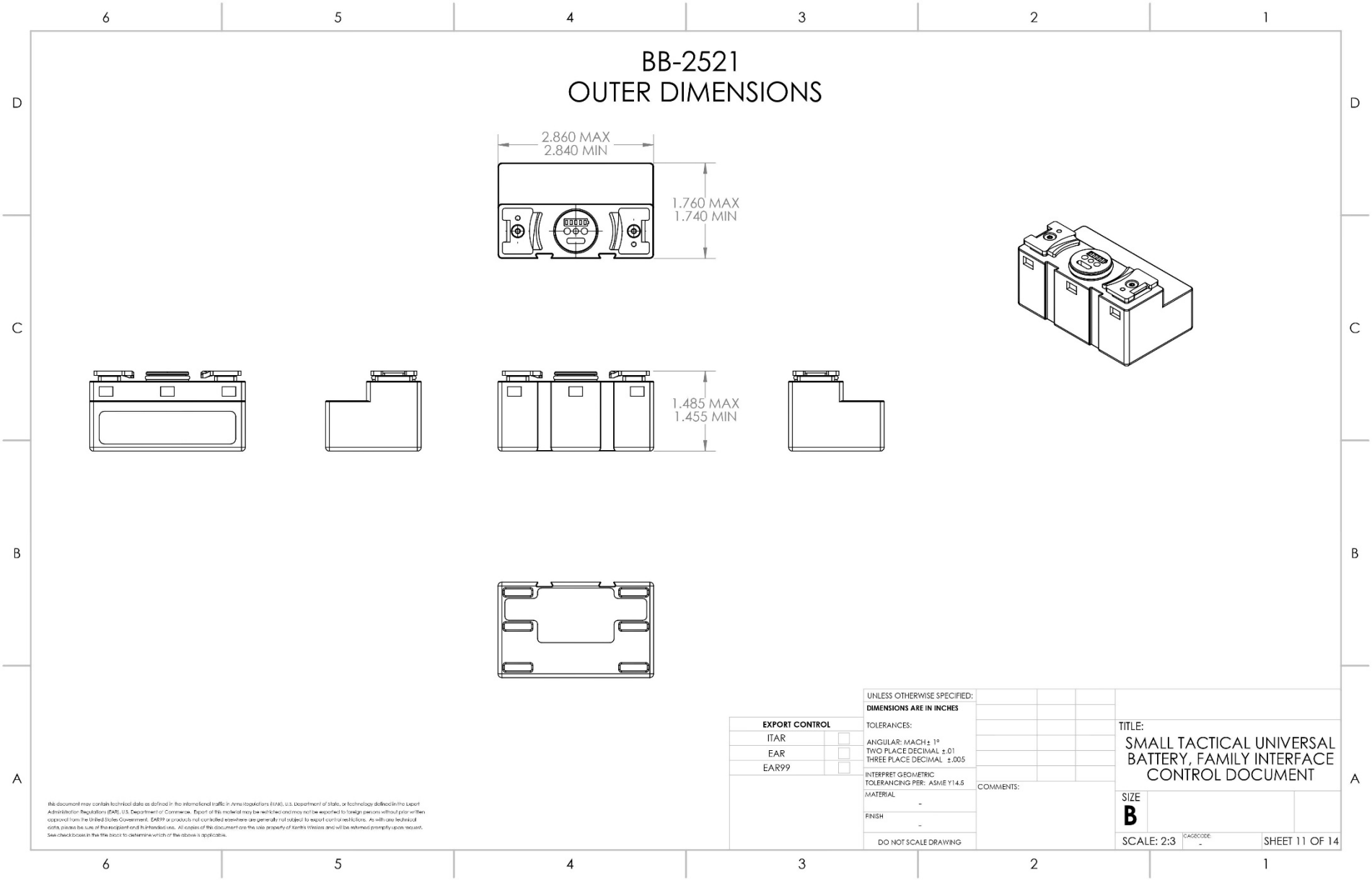
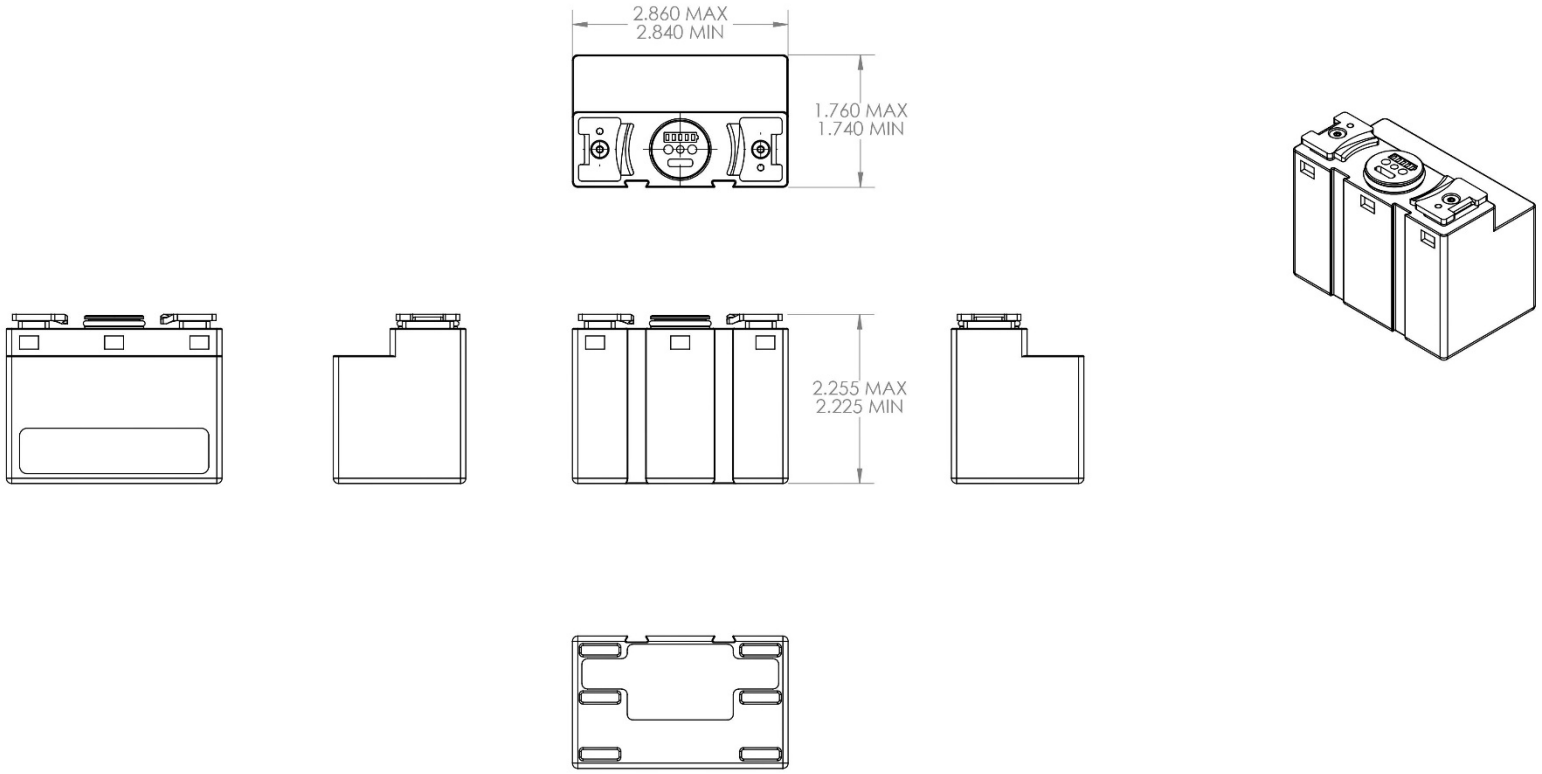


FIGURE 11. BB-2521/U

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THREE PLACE DECIMAL ±.005
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SCALE: 2:3
CAGECODE: -
SHEET 12 OF 14

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FIGURE 12. BB-2522/U

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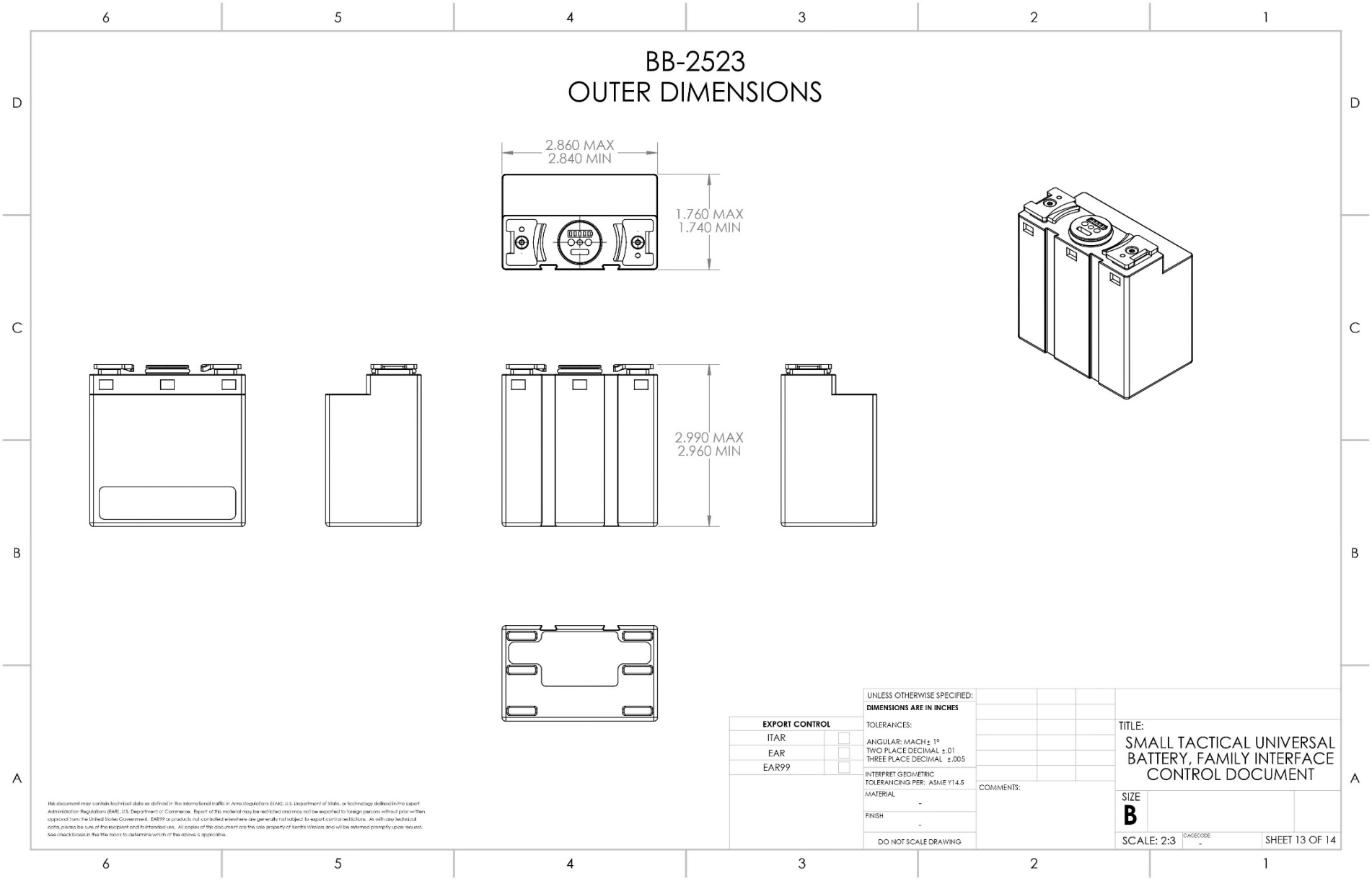


FIGURE 13. BB-2523/U

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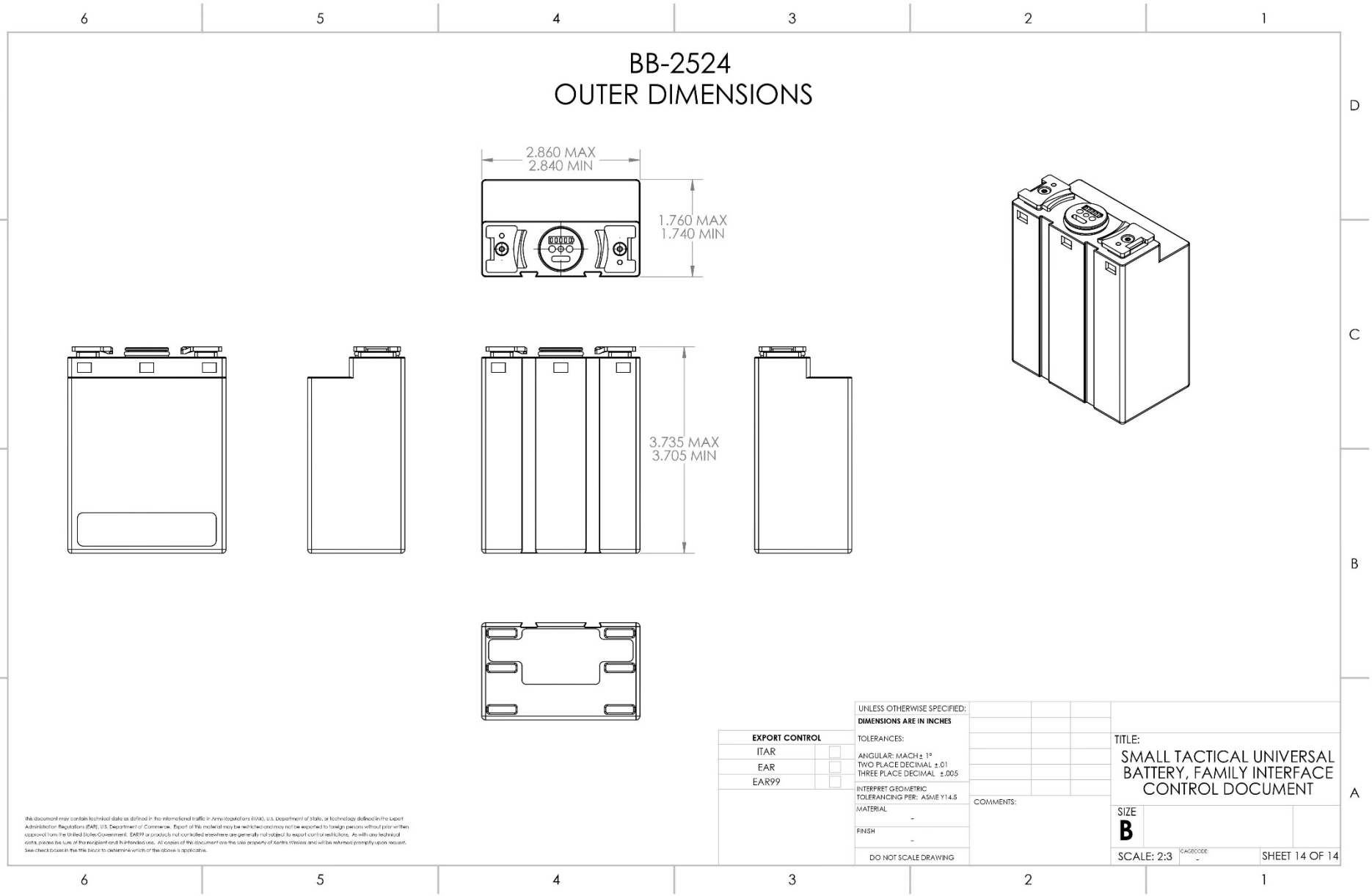


FIGURE 14. BB-2524/U

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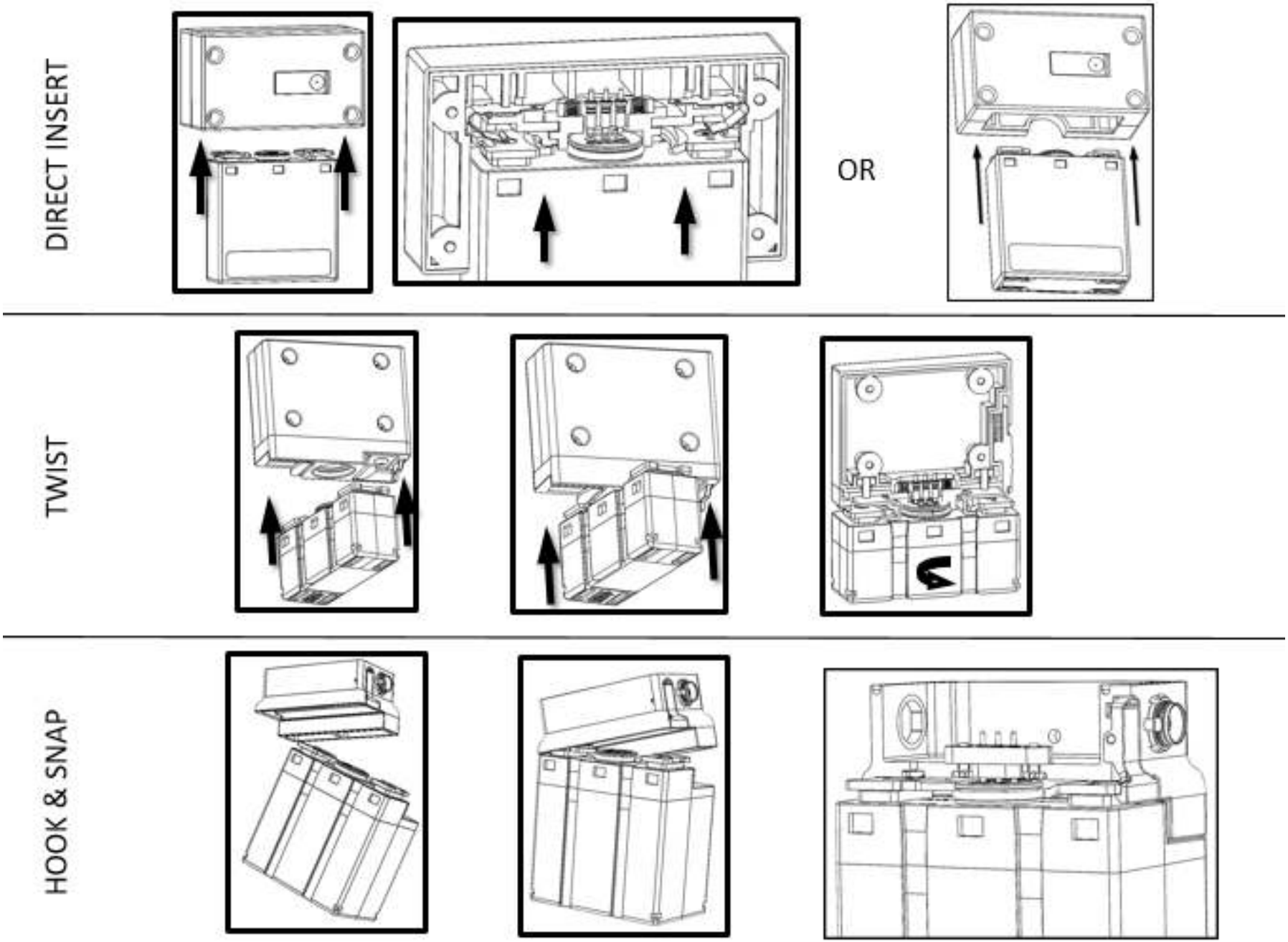


FIGURE 15. Methods of attachment

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NOTES: (This section contains information of a general nature which may be helpful, but is not mandatory)

Nominal ratings: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

Battery PIN: M32383/7	**-3**							
Type Classification:	BB-2511/U	BB-2512/U	BB-2513/U	BB-2514/U	BB-2521/U	BB-2522	BB-2523/U	BB-2524/U
Chemistry:	Li-Ion							
Color:	Tan 686A							
Weight (max):	0.27 lbs	0.39 lbs	0.52 lbs	0.64 lbs	0.39 lbs	0.64 lbs	0.89 lbs	1.20 lbs
Voltage Range:	3.3-21 V							
Energy (new battery):	11 Wh	22 Wh	32 Wh	43 Wh	22 Wh	43 Wh	65 Wh	86 Wh
Rated power delivery:	5V/3A 9V/1.67A 12V/1.25A 15V/1A 20V/0.75A 3.3-5.9V/3A 3.3-11V/1.5A	5V/3A 9V/2.75A 12V/2A 15V/1.5A 20V/1.125A 3.3-11V/3A 3.3-21V/1.5A	5V/3A 9V/3A 12V/3A 15V/2A 20V/1.5A 3.3-11V/3A 3.3-21V/2A	5V/3A 9V/3A 12V/3A 15V/3A 20V/2A 3.3-16V/4A 3.3-11V/2A	5V/3A 9V/2.75A 12V/2A 15V/1.5A 20V/1.125A 3.3-11V/3A 3.3-21V/1.5A	5V/3A 9V/3A 12V/3A 15V/3A 20V/2A 3.3-16V/4A 3.3-11V/2A	5V/5A 9V/5A 12V/5A 15V/4A 20V/3A 3.3-16V/4A 3.3-21V/3A	5V/5A 9V/5A 12V/5A 15V/5A 20V/4A 3.3-16V/5A 3.3-21V/4A
Battery Life:	≥ 224 cycles, ≥ 3 years							
Charge temperature range:	32 to 113ºF (0 to 45ºC)							
Operating temperature range:	-4 to 140ºF (-20 to 55ºC)							
Storage temperature range:	-20 to 60 C (1 month), -20 to 45 C (3 months), -20 to 20 C (12 months)							
MAX abusive temperature: (non-operating):	199ºF (93ºC)							
Connector:	Figure 2							
Overall Dimensions (inches):	2.85 x 1.00 x 1.47	2.85 x 1.00 x 2.24	2.85 x 1.00 x 2.97	2.85 x 1.00 x 3.72	2.85 x 1.75 x 1.47	2.85 x 1.75 x 2.24	2.85 x 1.75 x 2.97	2.85 x 1.75 x 3.72

Other data:

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for the BB-25XX()/U batteries during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

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Custodians:
Army – CR
Navy – MC
DLA - CC

Review Activities:
Navy – AS, SH
Air Force – 71

Preparing Activity:
Army – CR

(Project Number 6140-2021-018)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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